

NOTES AND DISCUSSIONS | MAY 01 2024

Which is greater: e^π or π^e ? An unorthodox physical solution to a classic puzzle ✓

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Which is greater: e^π or π^e ? An unorthodox physical solution to a classic puzzle

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The question of the title is a famous mathematical puzzle and can be addressed by several approaches. In this contribution, we present an alternative solution based on the second law of thermodynamics. The method can be extended to derive a more general result involving the exponential function. © 2024 Published under an exclusive license by American Association of Physics Teachers.

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Usually, we use mathematics to solve physics problems. This brief note provides an example of the reverse: A physical law (the second law of thermodynamics) is employed to solve a mathematical puzzle (the title of this paper).

This proof may surprise and interest students, motivating them to explore thermodynamic paths to generate other algebraic inequalities, which results in an improvement in their skills in performing entropy analyses.

The proposed solution is as follows. When two bodies A and B at different temperatures are placed in contact, energy flows between them until thermal equilibrium are reached. Let us suppose that A is an incompressible solid with constant heat capacity C , initially at temperature $T_1 = \pi$, and B is an ideal thermal reservoir at temperature $T_2 = e$, where e is Euler's number; both temperatures measured in the same absolute temperature scale.¹ We will assume that each system exchanges heat only with the other.

Once thermal equilibrium is reached, the entropy change of the solid is

$$\Delta S_A = C \ln \left(\frac{T_2}{T_1} \right) = C(1 - \ln \pi). \quad (1)$$

The reservoir absorbs all the heat released by the solid

$$Q_B = -Q_A = -\Delta U_A = C(T_1 - T_2) = C(\pi - e), \quad (2)$$

so the entropy of the reservoir changes in the amount

$$\Delta S_B = \frac{Q_B}{T_2} = C \left(\frac{\pi}{e} - 1 \right). \quad (3)$$

Finally, from Eqs. (1) and (3), we have that the global entropy change associated with the thermalization process is

$$\Delta S_{universe} = \Delta S_A + \Delta S_B = C \left(\frac{\pi}{e} - \ln \pi \right). \quad (4)$$

The key point is that, since the initial temperature of the solid is different from the reservoir's temperature, the heat transfer is irreversible and, consequently, the total entropy change must be positive. Because $C > 0$, Eq. (4) yields

$$\frac{\pi}{e} - \ln \pi > 0 \Rightarrow \pi > \ln \pi^e \Rightarrow e^\pi > \pi^e, \quad (5)$$

so we conclude that e^π is the greater of the two numbers.² This is known as *Gelfond's constant*, and it is a transcendental number whose approximate value is $e^\pi \simeq 23.14069$. On the other hand, $\pi^e \simeq 22.45915$, and it is unknown whether or not it is transcendental.³

An alternative derivation involves a perfect gas in thermal equilibrium with a reservoir. If the initial pressure is $P_1 = e$ and we suddenly increase the external pressure to the constant value $P_2 = \pi$,⁴ after transients have died out, the system will reach a new equilibrium state with the bath at pressure P_2 and at the initial temperature. Using the fact that the entropy change of the gas is $R \ln (P_1/P_2)$, where R is the gas constant, and noting that the heat exchange equals the work done on the gas, we obtain that the entropy change of the universe is

$$\Delta S_{universe} = R \left(\frac{\pi}{e} - \ln \pi \right), \quad (6)$$

from which the same conclusion is derived. It is also interesting to note that if the final pressure (or, in the previous example, the initial temperature) is some arbitrary positive value x , analogous reasoning allows us to infer the following more general inequality:

$$e^x \geq x^e, \quad x \geq 0. \quad (7)$$

Since the results obtained are mathematical truths, and, therefore, independent of any physical considerations, it is interesting to discuss in the classroom whether the thermodynamic derivation is, in fact, a proof in the mathematical sense. We refer the readers to Refs. 5–7 to delve deeper into this issue.

Instructors interested in this approach can find thermodynamic derivations of other famous inequalities in Refs. 8–10 (inequalities between means), Ref. 11 (Jensen's inequality), or Ref. 12 (Bernoulli's inequalities and bounds for the logarithmic function).

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

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¹Although thermometers do not have infinite precision, using $e \simeq 2.7$ and $\pi \simeq 3.1$ is enough to obtain the correct relation between the exponentials in the title.

²A compilation of solutions, some of them very ingenious, can be found in <https://mindyourdecisions.com/blog/2013/08/05/monday-puzzle-what-is-greater-epi-or-pie/> (visited on 11/21/2023). The reader may verify that if the temperatures of the solid and the reservoir are exchanged, the following tighter result is obtained: $\ln \pi + e/\pi > 2$ (the relative difference between the terms is less than 0.5%).

³C. A. Pickover, *Pickover, Wonders of Numbers* (Oxford U. P., Oxford, 2001).

⁴Note that, since the process is out of equilibrium, the pressure exerted on the gas is different from the gas pressure, which is possibly not even well-defined during the whole process.

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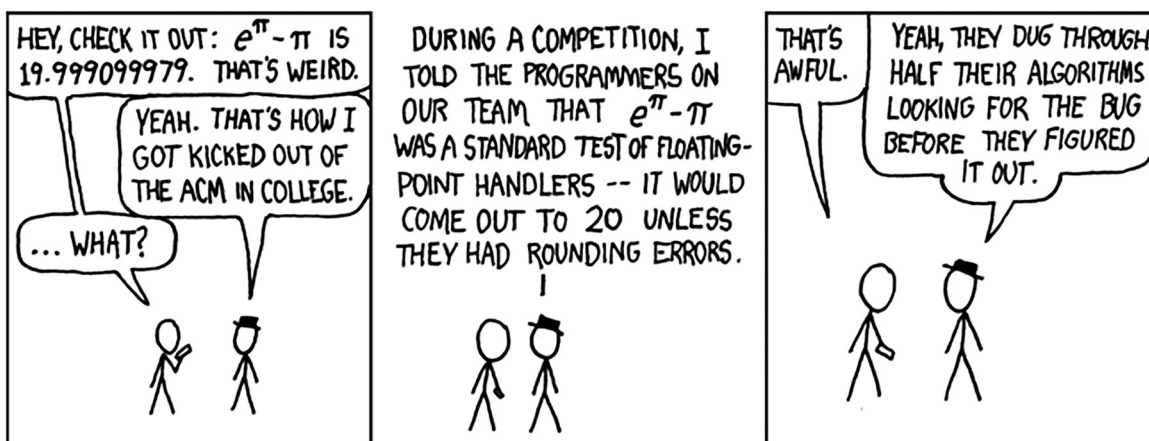
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Also, I hear the 4th root of $(9^2 + 19^2/22)$ is pi. (Source: <https://xkcd.com/217>)